Hao Sun

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RESEARCH INTERESTS

My research makes OS kernels correct and secure: **safe-by-construction**, via proof-guided in-kernel verifiers that admit only mechanically verified extensions; and **correct-by-construction**, uniting coding agents with formal verification to generate code with machine-checked proofs. In parallel, I expose deep kernel defects at scale through automated testing and fuzzing.

Keywords: *Operating Systems, Program Verification, Formal Methods, Kernel Fuzzing*

EDUCATION

ETH Zurich — *2023 – present*
Ph.D. Student, Advanced Software Technologies (AST) Lab — *Advisor: Prof. Zhendong Su*

Tsinghua University — *2020 – 2023*
Master of Science in Software Engineering — *Advisor: Prof. Yu Jiang*

Beijing University of Posts and Telecommunications — *2016 – 2020*
Bachelor of Science in Software Engineering

EXPERIENCE

Microsoft Research, Redmond, USA — *2026*
Research Intern — *Mentors: Jay Lorch, Ziqiao Zhou*
- A gallery of *verified* OS kernel components in Rust using Verus.
- Starting with a Linux-quality, formally verified concurrent page table.

City University of Hong Kong, Hong Kong — *2022*
Research Assistant (6 months) — *Mentor: Prof. Nan Guan*
- Security of the ROS 2 (Robot Operating System) robotic software stack.

AWARDS & HONORS

Google PhD Fellowship, ETH Zurich CS Department nominee (*under review by Google*)	*2026*
Program Committee, 4th Workshop on eBPF and Kernel Extensions (co-located with SOSP '26)	*2026*
Invited talk, ETH Zurich Systems Group Seminar	*2026*
SOSP '25 Best Paper Award	*2025*
eBPF Foundation Research Grant ($50,000)	*2024*
Invited talk, Linux Plumbers Conference (LPC '24), eBPF Track	*2024*
Tsinghua Outstanding Master's Graduate (top graduate honor)	*2023*
National Scholarship, Tsinghua University	*2022*

OPEN-SOURCE IMPACT ON LINUX

- SOSP '25 work shared as six patchsets, driving significant verifier changes.
- OSDI '24 methodology adopted by the community; led to algorithm-level verifier fix.
- Novel 4-line, formally-verified algorithm replacing a 27-line routine, merged upstream.
- 50+ patches merged, incl. exploitable verifier bug fixes and new kernel primitives.

PUBLICATIONS *(Google Scholar)*

348 citations (June 2026). Underlined name indicates the author.

eBPF verifier and kernel correctness:

From Specification to Kernel Commit: Verified Code Generation on Real-World Systems.
Hao Sun, Zenan Li, Cong Li, and Zhendong Su.
Under submission to ASPLOS '27.

Fast and Precise In-Kernel Extension Analysis via Proof-Guided Abstraction Refinement.
Hao Sun and Zhendong Su.
Invited paper*; under review at ACM Transactions on Computer Systems (TOCS), 2026.*

Prove It to the Kernel: Precise Extension Analysis via Proof-Guided Abstraction Refinement.
Hao Sun and Zhendong Su.
In Proceedings of SOSP, Seoul, Korea, October 13-16, 2025.
🏆 Best Paper Award

Approximation Enforced Execution of Untrusted Linux Kernel Extensions.
Hao Sun and Zhendong Su.
In Proceedings of USENIX Security, Seattle, WA, USA, August 13-15, 2025.

Validating the eBPF Verifier via State Embedding.
Hao Sun and Zhendong Su.
In Proceedings of OSDI, Santa Clara, CA, USA, July 10-12, 2024.
🐧 Methodology adopted by the Linux kernel community

Finding Correctness Bugs in eBPF Verifier with Structured and Sanitized Program.
Hao Sun, Yiru Xu, Jianzhong Liu, Yuheng Shen, Nan Guan, and Yu Jiang.
In Proceedings of EuroSys '24.

Kernel fuzzing:

KSG: Augmenting Kernel Fuzzing with System Call Specification Generation.
Hao Sun, Yuheng Shen, Jianzhong Liu, Yiru Xu, and Yu Jiang.
In Proceedings of USENIX ATC '22.

HEALER: Relation Learning Guided Kernel Fuzzing.
Hao Sun, Yuheng Shen, Cong Wang, Jianzhong Liu, Yu Jiang, Ting Chen, and Aiguo Cui.
In Proceedings of SOSP '21.

Co-authored publications:

Stratified Testing: Divide the Input Space, Conquer Rare Paths.
Zijing Yin, **Hao Sun**, David Basin, and Zhendong Su.
Under submission to USENIX ATC '26.

BPF-Ext: Safely Extending the eBPF Compilation Pipeline with Native Operations.
Yusheng Zheng, Zhengjie Ji, Weichen Tao, **Hao Sun**, Wei Zhang, Dan Williams, and Andi Quinn.
Under submission to USENIX ATC '26.

Synsema: Syntax-Guided Learning of Semantically Valid Programs.
Levin Winter, Cong Li, **Hao Sun**, and Zhendong Su.
Under submission to NeurIPS '26.

Optimizing Input Minimization in Kernel Fuzzing.
Hui Guo, **Hao Sun**, Shan Huang, Ting Su, Geguang Pu, and Shaohua Li.
In Proceedings of USENIX ATC 2025.

Finding Metadata Inconsistencies in Distributed File Systems via Cross-Node Operation Modeling.
Fuchen Ma, Yuanliang Chen, Yuanhang Zhou, Zhen Yan, **Hao Sun**, and Yu Jiang.
In Proceedings of USENIX Security 2025.

SnapCC: Effective File System Consistency Testing Using Systematic State Exploration.
Jianzhong Liu, Yuheng Shen, Yiru Xu, **Hao Sun**, and Yu Jiang.
ACM Transactions on Software Engineering and Methodology 2025.

SATURN: Host-Gadget Synergistic USB Driver Fuzzing.
Yiru Xu, **Hao Sun**, Jianzhong Liu, Yuheng Shen, and Yu Jiang.
IEEE Symposium on Security and Privacy (S&P '24)

Enhancing ROS System Fuzzing through Callback Tracing.
Yuheng Shen, Jianzhong Liu, Yiru Xu, **Hao Sun**, Mingzhe Wang, Nan Guan, Heyuan Shi, and Yu Jiang.
In Proceedings of ISSTA '24.

Effectively Sanitizing Embedded Operating Systems.
Jianzhong Liu, Yuheng Shen, Yiru Xu, **Hao Sun**, Heyuan Shi, and Yu Jiang.
In Proceedings of ACM/IEEE Design Automation Conference (DAC '24).

Horus: Accelerating Kernel Fuzzing Through Efficient Host-VM Memory Access Procedures.
Jianzhong Liu, Yuheng Shen, Yiru Xu, **Hao Sun**, and Yu Jiang.
ACM Transactions on Software Engineering and Methodology 2023.

Tardis: Coverage-Guided Embedded Operating System Fuzzing.
Yuheng Shen, Yiru Xu, **Hao Sun**, Jianzhong Liu, Zichen Xu, Aiguo Cui, Heyuan Shi, and Yu Jiang.
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (EMSOFT '22).

Rtkaller: State-Aware Task Generation for RTOS Fuzzing.
Yuheng Shen, **Hao Sun**, Yu Jiang, Heyuan Shi, Yixiao Yang, and Wanli Chang.
ACM Transactions on Embedded Computing Systems (EMSOFT '21).

Go-Sanitizer: Bug-Oriented Assertion Generation for Golang.
Cong Wang, **Hao Sun**, Yiwen Xu, Yu Jiang, Huafeng Zhang, and Ming Gu.
In Proceedings of IEEE Int'l Symposium on Software Reliability Engineering (ISSRE '19).

SELECTED PROJECTS

BCF: eBPF Certificate Framework for Proof-Guided Abstraction Refinement

- Significantly enhance the eBPF verifier while preserving *linear-time* complexity in the kernel.
- A from-scratch proof checker ($\sim$5K LoC in C) validates binary certificates with low overhead — $10\times$–$65.5\times$ faster than its user-space counterpart.
- **Impact**: SOSP '25 Best Paper; working on upstreaming the project to Linux.

Vero: A Real-World Benchmark for Kernel-Grade Verified Code Generation

- Verified-code-generation tasks drawn from the Linux eBPF verifier, LLVM, and the seL4 microkernel; each task encodes a *new* specification with no public reference solution.
- **Impact**: agent-derived, machine-checked algorithms upstreamed to the kernel stack (a branchless 4-line `tnum_step` merged into Linux); LLVM `KnownBits` patch in preparation.

Healer: Relation Learning Guided Kernel Fuzzing

- System call influence relation learning for boosting fuzzing coverage and crash discovery.
- A from-scratch, highly efficient kernel fuzzer written in Rust ($\sim$17K LoC).
- **Impact**: 290+ GitHub stars; 100+ Linux bugs found and fixed; 10+ CVEs; SOSP '21 artifact.

TEACHING EXPERIENCE

Stochastics and Machine Learning, ETH Zurich *2026*
Teaching Assistant

Compiler Design, ETH Zurich *2023, 2024, 2025*
Teaching Assistant

Data Modeling and Database (DMDB), ETH Zurich *2024, 2025*
Teaching Assistant

Software Engineering Seminar, ETH Zurich *2024, 2025*
Teaching Assistant

STUDENT SUPERVISION

Tynan Richards, Master Student, ETH Zurich *Master's Thesis, 2026*
Reentrancy-safe smart contract language via compile-time effect system.

Hoang Duong, École Polytechnique *SSRF Internship, 07/2025 – 09/2025*
eBPF JIT compiler benchmark; discovered in-kernel JIT optimization opportunities (to be submitted to the eBPF Workshop '26).

Li Shi, Master Student, ETH Zurich *Master's Thesis, 2025*
Hunting io_uring bugs through context-free workload generation.

Andreas Bur, Master Student, ETH Zurich *Practical Work, 2024*
Building a fuzzer specifically for the KVM subsystem in Linux for security bugs.

Jason Zibung, Master Student, ETH Zurich *Practical Work, 2024*
Low-level system design; snapshotting multi-threaded programs.